QUANTUM MECHANICS

1. Introduction

Quantum Mechanics is a branch of physics that deals with the behavior of matter and energy at very small scales, such as atoms, electrons, photons, and other microscopic particles. Classical mechanics, developed mainly by Newton, successfully explains the motion of everyday objects, but it cannot properly explain many phenomena occurring at the atomic and subatomic levels.

Quantum mechanics introduced a new way of understanding nature in which physical quantities such as energy, momentum, and position are described using probabilities and wave functions.

Some important phenomena that led to the development of quantum mechanics are:

- Black body radiation
- Photoelectric effect
- Atomic spectra
- Compton effect
- Wave-particle duality
- Stability of atoms

Quantum mechanics has applications in modern technologies such as lasers, semiconductors, transistors, quantum computers, medical imaging, and electron microscopes.

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2. Wave-Particle Duality

One of the most important concepts of quantum mechanics is that microscopic objects can behave both as particles and as waves.

Particle Nature of Light

Light was traditionally considered to be an electromagnetic wave. However, experiments such as the photoelectric effect showed that light also behaves as a collection of particles called **photons**.

The energy of a photon is given by:

E = hv

Where:

- E = Energy of photon
- h = Planck's constant
- v = Frequency of radiation

The momentum of a photon is:

$p = h/\lambda$

Where λ is the wavelength of light.

Wave Nature of Matter

In 1924, Louis de Broglie proposed that if light can behave as both a wave and a particle, then particles such as electrons should also have wave-like properties.

According to de Broglie:

$\lambda = h/p$

Since momentum p = mv,

$\lambda = h/mv$

This wavelength is called the **de Broglie wavelength**.

Thus, every moving particle is associated with a wave. This concept is known as **matter waves**.

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3. Heisenberg's Uncertainty Principle

The Heisenberg Uncertainty Principle was proposed by Werner Heisenberg.

It states that:

It is impossible to determine simultaneously and accurately both the position and momentum of a microscopic particle.

Mathematically,

$\Delta x \, \Delta p \geq \hbar/2$

Where:

- Δx = uncertainty in position
- Δp = uncertainty in momentum
- $\hbar$ = h/2π

Since p = mv, the principle can also be written as:

$\Delta x \, \Delta v \geq \hbar/2m$

Significance

The uncertainty principle is not simply a limitation of measuring instruments. It is a fundamental property of microscopic particles.

For example, if the position of an electron is determined very accurately, its momentum becomes more uncertain. Similarly, if its momentum is determined accurately, its position becomes uncertain.

This principle is one of the major differences between classical and quantum mechanics.

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4. Wave Function

In quantum mechanics, the state of a particle is represented by a mathematical function called the **wave function**, represented by the symbol:

ψ (psi)

The wave function contains information about the quantum state of a particle.

The wave function itself does not directly represent a measurable physical quantity. However, according to Max Born's interpretation, the square of the magnitude of the wave function represents the probability density of finding the particle.

Therefore:

Probability density = $|\psi|^2$

For a particle to exist somewhere in space, the wave function must satisfy certain conditions.

A physically acceptable wave function should generally be:

- Finite
- Single-valued
- Continuous
- Normalizable

Normalization

The total probability of finding a particle somewhere in space must be equal to 1.

Therefore:

$$\int |\psi|^2 \, dV = 1$$

This is known as the **normalization condition**.

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5. Schrödinger Wave Equation

The Schrödinger equation is one of the fundamental equations of quantum mechanics. It was developed by Erwin Schrödinger in 1926.

It describes how the quantum state of a physical system changes with time.

Time-Dependent Schrödinger Equation

The general time-dependent Schrödinger equation is:

$$i\hbar \, \partial\psi/\partial t = \hat{H}\psi$$

Where:

- i = imaginary unit
- $\hbar$ = reduced Planck's constant
- ψ = wave function
- t = time
- $\hat{H}$ = Hamiltonian operator

For a particle moving in a potential field V, the equation can be written as:

$$i\hbar \, \partial\psi/\partial t = [-\hbar^2/2m \, \nabla^2 + V]\psi$$

This equation is used to describe the evolution of a quantum system with time.

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6. Time-Independent Schrödinger Equation

For systems where the potential energy does not depend explicitly on time, the Schrödinger equation can be simplified.

The time-independent Schrödinger equation is:

$$\hat{H}\psi = E\psi$$

or

$$[-\hbar^2/2m \, \nabla^2 + V]\psi = E\psi$$

Where:

- E = Total energy of the particle
- V = Potential energy
- ψ = Wave function

The equation is particularly important for finding the allowed energy levels of quantum systems.

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7. Particle in a One-Dimensional Box

The particle-in-a-box model is one of the simplest and most important examples of quantum mechanics.

Consider a particle confined inside a one-dimensional box of length L.

The particle cannot exist outside the box. Therefore:

V = 0, for 0 < x < L

and

V = ∞, outside the box.

The allowed energy levels are:

$$\mathbf{E_n = n^2h^2 / 8mL^2}$$

where:

- n = 1, 2, 3, ...
- h = Planck's constant
- m = mass of particle
- L = length of box

The energy is therefore **quantized**, meaning that the particle can have only certain discrete energy values.

For n = 1:

$$\mathbf{E_1 = h^2 / 8mL^2}$$

This is called the **ground-state energy**.

For n = 2:

$$\mathbf{E_2 = 4h^2 / 8mL^2}$$

Thus:

$E_2 = 4E_1$

This shows that the energy of a quantum particle does not vary continuously.

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8. Quantum Numbers

Quantum numbers are used to describe the state of an electron in an atom.

There are four important quantum numbers:

1. Principal Quantum Number (n)

It represents the main energy level or shell of an electron.

Possible values are:

n = 1, 2, 3, ...

As n increases, the energy and average distance of the electron from the nucleus generally increase.

2. Azimuthal Quantum Number (l)

It describes the subshell and shape of the orbital.

Possible values are:

l = 0, 1, 2, ..., n − 1

The values correspond to:

- l = 0 → s orbital
- l = 1 → p orbital
- l = 2 → d orbital
- l = 3 → f orbital

3. Magnetic Quantum Number (m_l)

It describes the orientation of an orbital in space.

Its values range from:

−l to +l

4. Spin Quantum Number (m_s)

It describes the spin state of an electron.

The possible values are:

m_s = +1/2 or −1/2

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9. Applications of Quantum Mechanics

Quantum mechanics has become an essential part of modern science and technology.

Semiconductors

The operation of transistors, diodes, and integrated circuits depends on quantum mechanical principles. Modern computers and smartphones are based on semiconductor technology.

Lasers

LASER stands for **Light Amplification by Stimulated Emission of Radiation**.

Quantum mechanics explains the interaction between atoms and photons that makes laser operation possible.

Lasers are used in:

- Communication
- Medicine
- Manufacturing
- Barcode scanners
- Optical storage

Electron Microscopes

Electron microscopes use the wave nature of electrons to obtain images of extremely small objects. They can provide much greater magnification than ordinary optical microscopes.

Quantum Computing

Quantum computing uses quantum phenomena such as **superposition** and **entanglement** to process information.

Unlike classical bits, which can have a value of 0 or 1, a quantum bit or **qubit** can exist in a combination of states.

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10. Conclusion

Quantum mechanics provides a mathematical framework for understanding the behavior of matter and energy at microscopic scales. Important concepts such as wave-particle duality, uncertainty principle, wave function, Schrödinger equation, and quantization form the foundation of quantum theory.

Although quantum mechanics differs significantly from classical mechanics, it has been extremely successful in explaining experimental observations and predicting the behavior of microscopic systems.

Today, quantum mechanics is not only a fundamental theory of physics but also the basis of many modern technologies, including semiconductors, lasers, medical instruments, and emerging quantum computers.